Chapter 36
Energy, Commodity, and Environmental Derivatives

Section 5.8 priced a commodity forward with storage costs and a convenience yield, and Section 5.14 left to this chapter the markets where the term structure carries the most information. What makes it carry information is a single asymmetry. In equity and rates, the forward curve is derived from the spot by arbitrage; a violation is a trade. In commodities the arbitrage runs one way only—you can buy and store, but you cannot short and un-store—so the curve carries information about physical conditions that no arbitrage forces into line.

That asymmetry generates everything in this chapter. The curve's shape means something, seasonality survives, volatility depends on maturity, and one commodity, electricity, breaks the framework entirely because it cannot be stored at all.

The chapter then follows the asymmetry to its limit. Emissions allowances are storable at no cost, so the carry relation binds tighter than for any physical commodity and its residual means something else; allowance options have a bimodal density and a smile that is not a smile; and weather and catastrophe contracts have an underlying that cannot be held, so the writer cannot hedge and the price is an expectation rather than a replication cost. Those last sections show how much of Parts II and III depended on a hedge existing.

Sections 36.1 to 36.3 read the curve: what its shape says about inventories, why seasonality survives, and why volatility depends on maturity. Sections 36.4 to 36.8 take electricity and the contracts written on it, the spreads that turn one commodity into another, storage as an option, and the models these markets need; Sections 36.9 to 36.12 cover basis, the stack-and-roll hedge, financial participation, and the remaining markets. Sections 36.13 to 36.16 follow the asymmetry to its limit through emissions, weather, and catastrophe risk, and Section 36.17 sorts what is priced by arbitrage from what is priced by expectation. Section 36.18 says who trades these markets and why, and Section 36.19 how the chapter's results were obtained.

36.1 THE CURVE IS THE INFORMATION

Section 5.8's relation is

$$F(t,T) = S(t)\, e^{(r+u-y)(T-t)} \tag{36.1}$$

with u the storage cost and y the convenience yield, the benefit of holding the physical commodity rather than a claim on it.

The convenience yield is not observed: it is the residual that makes the equation hold, and treating it as a parameter with an independent existence is the most common error in commodity finance. But the residual is informative. A high convenience yield (a market in **backwardation**) says physical inventory is scarce and someone will pay to hold it now. A low one (**contango**) says inventory is plentiful and the curve is compensating storage.

Two consequences follow that have no analog in financial markets.

The upper bound on the curve is enforced and the lower bound is not. If F rises far above (36.1), anyone can buy spot, store, and sell forward. If F falls below it, the arbitrage requires

borrowing the physical commodity, and for most commodities that market is thin or nonexistent. So contango is capped by storage economics and backwardation is not capped at all, which is why the deepest backwardations occur in genuine shortages and can be extreme.

And storage capacity is finite. When it fills, the cap disappears: the forward can trade above full-carry because there is nowhere to put the commodity. The most-quoted case is the April 2020 crude oil episode, when a front-month contract settled below zero because holders faced physical delivery with no storage available. That was not a market failure but the arbitrage bound behaving as it must when the constraint binds, and the clearest modern demonstration that a commodity futures contract is a claim on a physical thing.

Example 36.1 Reading inventory off the curve

Crude spot \$78.00, financing 4.5%, storage 6% of value per year. Inverting (36.1),

$$y = r + u - \frac{1}{T}\ln\frac{F}{S}$$

	Backwardated		Contango	
Months	F	**Implied** y	F	**Implied** y
1	77.40	19.77%	78.35	5.13%
3	76.30	19.31%	79.10	4.90%
6	75.00	18.34%	80.20	4.94%
12	73.20	16.85%	82.00	5.50%
24	71.50	14.85%	84.50	6.50%

In backwardation the implied yield far exceeds $r + u$, which is the market paying nearly 20% a year for the privilege of holding a physical barrel now.

The term structure of y is itself a forecast: it declines with maturity in the backwardated case, which says the shortage is expected to resolve, where a flat implied-yield curve would say the tightness is expected to persist.

A negative implied convenience yield is the diagnostic for full storage. It means the curve is steeper than full carry, which cannot be arbitraged away because there is nowhere to put the commodity. That steepness is the precondition for the April 2020 episode, but not the episode itself: (36.1) makes F a positive multiple of S, so no convenience yield, however negative, can take a futures price below zero. What went negative was the spot value of a barrel at the delivery point. Holders of the expiring contract had to take delivery with no tank to put it in, so owning a barrel had become a disposal cost, and on April 20, 2020 the May contract settled at −\$37.63 while June, a month later, settled at +\$20.43.

36.2 SEASONALITY

Financial forward curves are smooth; commodity curves are not. Natural gas, power, and agricultural curves have a stable annual shape driven by consumption:

Month	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Gas forward (\$/MMBtu)	2.85	2.88	2.95	3.05	3.08	3.02	3.10	3.45	3.80	3.95	3.85	3.30

The winter premium is not a forecast that prices will rise. It is the cost of the storage cycle: gas injected in summer and withdrawn in winter must earn the storage cost, the financing, and a return on the facility.

Two practical consequences follow.

A "roll yield" computed across a seasonal boundary is meaningless. Rolling from December into January in gas is not a term-structure trade; it is a seasonal step. Section 5.11's roll analysis applies within a season and not across one.

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And seasonality must be removed before a volatility is estimated: a series that rises every winter has a partly deterministic variance, and treating it as risk overstates the volatility. This is Section 33.10's point about inflation, with a larger amplitude.

36.3 THE SAMUELSON MATURITY EFFECT

The **Samuelson maturity effect**, named for Paul Samuelson, who predicted it in 1965, is the tendency of a futures contract's volatility to rise as its expiration approaches. Seen across the curve on any one day, it means near contracts are more volatile than far ones. The reason is mean reversion in the spot.

Example 36.2 Volatility falls with maturity

If the spot mean-reverts at rate a, a shock decays before a distant contract expires, so the futures volatility is

$$\sigma_F(T) = \sigma_S\, e^{-a(T-t)}, \tag{36.2}$$

with σ_S the volatility of the spot price and $T - t$ the contract's time to expiration. [TN36] derives it from a mean-reverting log spot price. The decay is the one Section 31.6 found for Hull–White forward rates, and for the same reason: mean reversion damps a shock before the distant date arrives. The quantities differ, though. There the short rate mean-reverts and the result is a basis-point volatility of a forward rate; here the logarithm of the spot price mean-reverts and the result is a percentage volatility of a futures price. At a 60% spot volatility:

Months out	$a = 0.5$	$a = 1.5$	$a = 3.0$
1	57.6%	52.9%	46.7%
3	52.9%	41.2%	28.3%
6	46.7%	28.3%	13.4%
12	36.4%	**13.4%**	3.0%
24	22.1%	3.0%	0.1%

At $a = 1.5$ the one-month contract carries 52.9% volatility and the twelve-month 13.4%, a factor of 3.96.

Three things follow.

Options on different contract months are not comparable at the same strike. A 40% implied volatility on the front month and on the one-year are describing very different markets, and a volatility surface for a commodity needs a maturity dimension that means something different from an equity surface's.

Hedge ratios across maturities are not one-for-one. Hedging a two-year exposure with front-month futures over-hedges the volatility by the ratio in the table, which is the commodity version of the duration mismatch that Section 6.7's duration-based hedge corrects.

And the effect competes with a second one. Storage links adjacent contracts, which *raises* correlation between them; mean reversion *lowers* the far contract's volatility. A commodity with cheap, ample storage has a flat volatility term structure and a highly correlated curve; one with costly or full storage has a steep one and a curve whose near and far ends can move independently. The volatility term structure is therefore a direct read on storage economics.

Figure 36.1 shows both effects: futures volatility falling with maturity, and a gas curve whose shape is set by the calendar rather than by the price level.

36.4 ELECTRICITY

Power cannot be stored, and that single fact removes most of this book's machinery.

(36.1) does not hold, because there is no cash-and-carry: you cannot buy power today and deliver it tomorrow. So the forward is not linked to the spot by arbitrage at all: it is an

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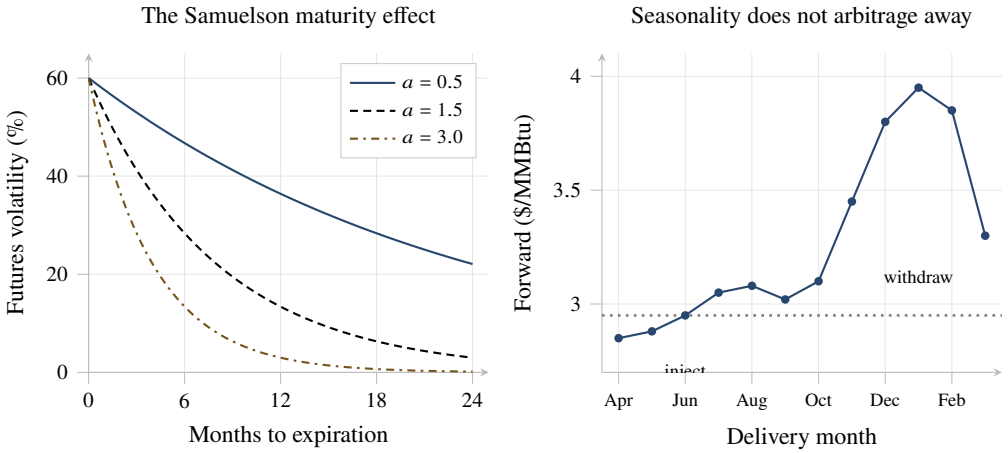


Figure 36.1: Left: equation (36.2) at $\sigma_S = 60\%$. Right: Section 36.2's gas curve.

expectation plus a risk premium, and nothing enforces it.

The price process is correspondingly unusual. Supply is a stack of generators ordered by marginal cost, and the price is set by the last unit needed. When demand approaches capacity the stack becomes near-vertical, so a small demand increase produces an enormous price move—and then reverts within hours when the weather changes.

A simple model captures this. On an ordinary day the price is lognormal; on a small fraction p of days demand reaches the steep end of the stack, and the day's price is multiplied by a random spike factor J:

$$P = m\, e^{\sigma Z} J^{B}, \qquad Z \sim N(0,1), \quad B \sim \text{Bernoulli}(p), \quad \ln J \sim N\big(\mu_J, s_J^2\big), \tag{36.3}$$

with Z, B and J independent. Here m is the median ordinary-day price (half of the days without a spike settle below it, and their mean is the slightly higher $m\, e^{\sigma^2/2}$), σ is the standard deviation of its logarithm, and $E[J] = e^{\mu_J + s_J^2/2}$ the average size of a spike. Call this the **spike model**. Because J is lognormal, the price on a spike day is lognormal too, so the price is a mixture of two lognormals: with probability $1-p$ its logarithm is normal with mean $\ln m$ and variance σ^2, and with probability p normal with mean $\ln m + \mu_J$ and variance $\sigma^2 + s_J^2$. Its mean is

$$E[P] = m\, e^{\sigma^2/2}\big(1 - p + p\, E[J]\big),$$

and its distribution function is the probability-weighted sum of the two lognormals' distribution functions. The q-th percentile x_q therefore solves

$$(1-p)\, N\!\left(\frac{\ln(x_q/m)}{\sigma}\right) + p\, N\!\left(\frac{\ln(x_q/m) - \mu_J}{\sqrt{\sigma^2 + s_J^2}}\right) = q,$$

which has no closed-form solution but has exactly one root, because the left side rises steadily from 0 to 1 as x_q increases, so a few steps of bisection find it.

A contract on the day's price is valued the same way, by conditioning on whether the day spikes. A daily cap struck at K, paying $\max(P-K,0)$, is worth

$$c(K) = (1-p)\, c^{\mathrm{B}}\big(F_0, K, \sigma\big) + p\, c^{\mathrm{B}}\Big(E[J]\, F_0,\ K,\ \sqrt{\sigma^2 + s_J^2}\Big), \qquad F_0 = m\, e^{\sigma^2/2}, \tag{36.4}$$

before discounting to the delivery date, where $c^{\mathrm{B}}(F,K,v) = F\,N(d_1) - K\,N(d_1 - v)$ with $d_1 = [\ln(F/K) + v^2/2]/v$ is Black's formula from Chapter 16 written with the total volatility

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v in place of $\sigma\sqrt{T}$. Each term is one regime: an ordinary day has mean F_0 and log standard deviation σ; a spike day has a mean $E[J]$ times larger, and the spike factor's own uncertainty adds to the variance. A floor is valued the same way with Black's put, and a forward on the day's price is worth $E[P]$. So every figure the model produces is exact.

The model has no time dimension. It describes the distribution of one day's price, with each day drawn independently and a spike lasting exactly one day: the "reverts within hours" above, taken to its limit. That is all a daily cap needs, because its payoff depends on one day's price and a strip of daily caps is worth the sum of its days. A payoff that depends on several days (a cap on a monthly average, a swing contract, a hedge carried through a spike) needs the path as well, and there the standard choice is a mean-reverting log price with jumps that decay within days, so that spikes arrive in clustered episodes rather than as independent days.

Example 36.3 Why a lognormal model misprices a price cap

Take the spike model (36.3) with an ordinary-day median $m = \$43$ and $\sigma = 30\%$. Spikes occur on 2.4% of delivery days ($p = 2.4\%$), about nine days in a 365-day year, with a spike factor of mean $E[J] = 5$ and $s_J = 60\%$. Against it, fit a single lognormal with the *same mean ($49.30) and the same standard deviation ($40.42)* and compare.

	Spike model	Matched lognormal
Mean	\$49.30	\$49.30
Median	\$43.39	\$38.12
99th percentile	\$206.81	\$202.15
99.9th percentile	\$573.80	\$349.60

Now price daily caps with (36.4). At a \$100 strike, $F_0 = 43\,e^{0.045} = \$44.98$, so an ordinary day is worth $c^{\mathrm{B}}(44.98, 100, 0.30) = 0.0238$; a spike day, with mean \$224.90 and total volatility $\sqrt{0.09 + 0.36} = 67.1\%$, is worth $c^{\mathrm{B}}(224.90, 100, 0.671) = 130.24$. Weighting by the probabilities,

$$c(100) = 0.976 \times 0.0238 + 0.024 \times 130.24 = 0.0232 + 3.1257 = 3.1490,$$

so 99% of the cap's value comes from the nine or so spike days a year. The matched lognormal is a single application of Black's formula, at the forward \$49.30 and the 71.7% volatility that reproduces the standard deviation of \$40.42. Across strikes:

Strike	Spike model	Lognormal	Ratio
\$ 60	5.3300	10.4989	0.51
\$100	3.1490	4.1344	0.76
\$200	1.6578	0.6518	**2.54**
\$400	0.5056	0.0483	**10.48**
\$800	0.0739	0.0016	46.70

Matched on mean and on volatility, the two models disagree by a factor of ten at a \$400 strike—and they disagree in opposite directions at the two ends.

The lognormal spends its variance on everyday fluctuation and the real distribution spends it on nine or so days a year. So the lognormal over-prices near-the-money caps, because it makes moderate moves too likely, and drastically under-prices far ones.

A utility buying a price cap is buying precisely that difference, and it is the single largest model risk in energy trading. No amount of calibrating a lognormal to at-the-money options fixes it. The lognormal already over-prices the cap near the money, so fitting that price lowers its volatility, and a lower volatility cheapens the far caps further.

Two further features of electricity have no financial-market analog.

Prices can be negative, and routinely are in systems with large inflexible or subsidized generation, because a generator may prefer to pay to keep running rather than shut down and

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Table 36.1: Electricity contracts, their users, and how each is priced.

Contract	What it pays	Main users	Priced by
Block futures and forwards	A fixed price for power over a month, quarter or year, baseload or peak hours; usually cash-settled on the average hourly spot price	Generators sell; retailers and large consumers buy	The market: the curve is an input
Swaps	The same, over the counter: fixed against the average spot price at a hub or zone	Utilities, industrial consumers	As forwards
Options on block futures	Calls and puts on a block forward	Generators and retailers shaping hedges	Black's formula with the block's own volatility
Daily and hourly options; caps and floors	One option per delivery day or hour on that period's spot price	Retailers capping cost; owners of peaking plants	A spot model with spikes, Example 36.3
Spark and dark spread options; tolling	Power less fuel; a tolling agreement rents a plant's output	Plant owners and buyers of capacity	Exchange option, Example 36.5; dynamic programming for operating constraints
Swing and load-following contracts	A fixed price with volume at the holder's choice, within daily and total limits	Retail suppliers whose load follows the weather	Dynamic programming or least-squares Monte Carlo
Financial transmission rights	The price difference between two nodes, the congestion rent	Generators and load-serving utilities hedging location	Auction prices and network simulation; no replication
Power purchase agreements	A fixed price for a plant's output as produced; a virtual agreement settles the difference in cash	Corporate buyers of renewable power; wind and solar developers	A forward at the output-weighted price, plus shape, volume and basis risk

restart. Any model requiring positivity, and any option formula written in log-space, fails outright, which is why power desks work in normal rather than lognormal terms, exactly as Section 29.8's rate markets did for the same reason. Neither model in Example 36.3 can produce one: both multiply positive factors, so the spike model repairs the upper tail and is silent about the lower. Its cap prices are unaffected, because a cap pays only in the upper tail. Where negative prices matter (a floor, or a generator with an obligation to keep running), the ordinary-day price has to be modeled additively, as a normal variable or a lognormal shifted below zero, with the spike added on top.

And the commodity is defined by location and by hour. Power at one node and at another is a different good, connected by transmission that can bind. The basis between locations is itself a traded market, and its behavior is closer to a congestion option than to a spread.

36.5 ELECTRICITY CONTRACTS AND HOW THEY ARE PRICED

Because nothing links the forward to the spot, an electricity market is a family of contracts, each written on a different slice of the same price (a delivery period, a day, an hour, a location), and each priced by its own method. Table 36.1 sets them out.

Three rules organize the last column.

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The forward curve is an input. Power cannot be carried, so (36.1) produces no forward price. A block forward is the expected average spot price over its delivery period under the pricing measure, and nothing arbitrages it toward today's spot. Desks therefore take the curve from the market and calibrate every other model to it, which reverses Chapter 5's order: there the spot and the cost of carry gave the forward, and here the forwards are the data.

The settlement period picks the model. A contract settled on an average over a month depends on how the whole delivery period moves and is priced with a model of the forward curve. A contract settled on one day or one hour depends on spikes and is priced with a model of the spot. A block forward averages hundreds of hourly prices, so its volatility is a fraction of the spot's, and the Samuelson maturity effect lowers it further the farther away delivery is. An option on a block is therefore cheap next to a strip of daily caps at the same strike. The next example shows by how much, and why the answer turns on something Example 36.3 could leave out.

Flexibility is a chain of linked options. The clearest case is the **swing contract**, a supply agreement at a fixed price in which the buyer chooses each day how much to take, between a daily minimum and maximum, subject to a minimum and maximum over the whole contract and a penalty for taking less than the total minimum. A gas retailer holding one takes the maximum on cold days, when spot gas costs more than the contract price, and the minimum on mild ones; what links the days is the total limit, because volume taken in December is not available in February. Tolling agreements and storage work the same way: the holder chooses how much, or when, on many dates, and each choice uses up capacity that a later one would have needed. Their value therefore depends on the whole price path rather than on one day's price, and it is found by dynamic programming over the price and the remaining volume, the method of Section 36.7, or by Chapter 20's least-squares Monte Carlo when the state is too large for a lattice.

Example 36.4 A daily cap and a monthly cap are different instruments

The example shows that two caps on the same price at the same strike can differ in value by a factor of eight, and that the gap depends on how spikes are spread through time, which the spike model of Example 36.3 does not specify. Take a 30-day month, a strike of \$60, and 1 MWh a day, with each day's price drawn from that example's spike model. A strip of daily caps pays $\sum_d \max(P_d - 60, 0)$; a cap on the monthly average pays $30\max(\bar{P} - 60, 0)$, with $\bar{P}$ the month's average price. Values are undiscounted.

The strip is worth $30 \times 5.3300 = \$159.90$, whatever the relation between days, because each of its terms depends on one day only. The average cap depends on the joint behavior of the days. The table varies that behavior while holding each day's distribution fixed: spikes arrive either on independent days or in episodes of three or five consecutive days, with the probability that a given day spikes kept at 2.4%. The average cap has no formula like (36.4), because its payoff depends on a sum of 30 prices, so it is valued by simulation: draw each day's ordinary price, whether the day spikes and its spike factor, average the month, and take the mean payoff over 400,000 simulated months for each row:

Spikes arrive	S.d. of $\bar{P}$	Monthly-average cap	Daily strip
On single days	\$ 7.41	\$19.3	\$159.90
In three-day episodes	\$10.11	\$44.3	\$159.90
In five-day episodes	\$12.15	\$65.9	\$159.90

Averaging removes most of a spike. A spike day that multiplies the price by five raises a 30-day average by only about \$6, so with independent days the monthly cap is worth an eighth of the strip. That is why options on block futures trade at volatilities far below those of daily options in the same market: the block's payoff is an average, and an average does not see a single day.

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Clustering raises the monthly cap and leaves the strip alone. Spikes in five-day episodes (a heat wave rather than a hot afternoon) more than triple the monthly cap, from \$19.3 to \$65.9. The dependence across days that Example 36.3 could ignore is the whole input to a monthly contract. A model calibrated to daily options says nothing about it, and a desk that prices monthly options as if days were independent will value them at less than half their worth if spikes in fact come in three-day episodes, and at less than a third if they come in five-day ones.

36.6 SPARK, DARK, AND CRACK SPREADS

The most important commodity derivatives are not on commodities but on the *margin between them*, because that is what an industrial owner actually earns.

Example 36.5 A power plant is an exchange option

A gas-fired generator earns $\max(P_e - \text{HR} \cdot P_g, 0)$ per MWh, where P_e is the power price per MWh, P_g the gas price per million British thermal units (MMBtu), and HR the **heat rate**, the MMBtu of gas the plant burns to produce one MWh. The margin $P_e - \text{HR} \cdot P_g$ is the **spark spread**. The plant runs when it is positive and shuts when it is not, so the plant *is* a strip of call options on the spark spread—and by Section 34.10 each is an **exchange option**: power against HR units of gas.

The example asks what one of those options is worth, and shows that the answer turns almost entirely on the correlation between power and gas. Take a peaking plant, one that runs only in the hours when power prices are high: power \$42.00/MWh, gas \$3.20/MMBtu, heat rate 12.5, six months, power volatility 55%, gas volatility 45%. Fuel cost is \$40.00/MWh, so the intrinsic spread is \$2.00. The option values below are forward values, undiscounted.

ρ	Effective vol	Option value	Time value
0.9	24.4%	\$3.929	\$1.929
0.7	39.8%	\$5.658	\$3.658
0.5	50.7%	\$6.891	\$4.891
0.3	59.7%	\$7.900	\$5.900
0.0	71.1%	\$9.171	\$7.171

Modeling power and gas as independent values the option on each megawatt-hour at \$9.171 instead of \$3.929, 133% too high.

Power and gas are strongly correlated because gas frequently sets the marginal price of power, so the spread is far less volatile than either leg. *High correlation makes a spread option cheap*, which is the opposite of the intuition that two volatile inputs must make a volatile output.

This is the single most consequential correlation assumption in energy finance, because it values the entire generation fleet: a merchant generator's enterprise value is the value of a portfolio of spark-spread options, and Section 34.10's warning that correlation must come from somewhere applies here with a whole plant's value at stake.

The same structure recurs. A **dark spread** is the coal-fired version. A **crack spread** is the refiner's:

Crude	\$78.00/bbl
Gasoline \$2.45/gal	\$102.90/bbl
Heating oil \$2.60/gal	\$109.20/bbl
3-2-1 crack $= (2 \times 102.90 + 109.20 - 3 \times 78.00)/3$	**\$27.00/bbl**

A refiner is long this and hedges by selling it: buying crude futures and selling product futures in the 3:2:1 ratio. The ratio is a technology statement, not a market convention: it approximates the yield of a typical US refinery, and a refiner with a different configuration hedging at 3-2-1 has a residual exposure to its own yield slate.

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36.7 STORAGE IS AN OPTION

A storage facility is the right, not the obligation, to buy at one date and sell at another. Its value therefore splits the way an option's does.

Example 36.6 Intrinsic and extrinsic

Using Section 36.2's curve, a facility can lock in a spread today: inject in the five cheapest months (April through September, skipping August, averaging \$2.95) and withdraw in the five most expensive (November through March, averaging \$3.67).

The intrinsic value is \$0.72 per MMBtu per cycle, and it can be hedged immediately with calendar spreads.

The extrinsic value is the right to change one's mind: to withdraw early if winter arrives early and cold, to re-inject if the spread widens. It is the time value of a strip of calendar-spread options, and it depends on volatility and on how correlated adjacent months are:

Monthly vol	$\rho = 0.90$	$\rho = 0.75$	$\rho = 0.50$
30%	0.2%	2.2%	7.1%
45%	1.7%	8.3%	17.4%
60%	5.1%	15.7%	27.0%
80%	11.0%	25.0%	**37.4%**

The entries are extrinsic value as a share of the total, with the strip valued as one exchange option (Margrabe's formula) on the \$3.67 withdrawal average against the \$2.95 injection average over six months, both at the volatility shown and correlation ρ between them.

At plausible parameters the extrinsic share is 10–30%, and it is the part a spreadsheet does not capture.

This explains a market structure fact. Storage capacity is often leased by trading firms rather than operated by the utility that built it, and the reason is not that traders are better at running compressors. A utility that books the intrinsic spread has sold the extrinsic value to whoever leases the capacity, because capturing it requires continuous re-optimization, which is an option-hedging operation, not a physical one.

And the same decomposition prices every physical asset with flexibility: a pipeline is a locational spread option, a swing contract on gas is a strip of daily options linked by its volume limits, and a liquefied natural gas (LNG) cargo with destination flexibility is an option on the basis between two markets.

36.8 MODELING

Four features have to be accommodated and no single model does all of them well.

Mean reversion generates Section 36.3's volatility term structure and is the minimum requirement.

Seasonality is handled deterministically (a seasonal function added to the log price) rather than stochastically, because it is predictable and modeling it as risk overstates volatility.

Use two factors, at least, for anything spread-related. A one-factor model makes all contract months perfectly correlated, which, once the model is fitted to a realistic curve, prices a calendar spread option at nearly zero. This is Section 31.12's defect in another market, and the fix is the same: a short-term mean-reverting factor plus a long-term factor with a random walk, which is the standard two-factor commodity model and produces both a decaying volatility term structure and imperfect correlation along the curve. Example 36.7 shows how much the second factor is worth.

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Example 36.7 Why a calendar spread needs two factors

The example shows that a one-factor model fitted to the same market as a two-factor model can value a calendar spread option at about a twelfth of the two-factor price. Let the market follow the two-factor model

$$\ln S = \chi + \xi, \qquad d\chi = -\kappa\chi\, dt + \sigma_\chi\, dW_\chi, \qquad d\xi = \mu\, dt + \sigma_\xi\, dW_\xi,$$

with independent shocks, a short-term factor at $\sigma_\chi = 40\%$ and $\kappa = 2$, and a long-term factor at $\sigma_\xi = 20\%$. The argument of [TN36], applied to each factor, gives the futures volatility at time to delivery τ as $\sigma_F(\tau) = \sqrt{\sigma_\chi^2 e^{-2\kappa\tau} + \sigma_\xi^2}$. Fit a one-factor model, $\sigma e^{-a\tau}$, to the one-year and two-year contracts, which is what a desk pricing a spread between them would do. Matching their 20.7% and 20.0% needs $a = 0.035$ and $\sigma = 21.5\%$:

Months to delivery	Two-factor (market)	One-factor fit
1	39.3%	21.4%
6	24.8%	21.1%
12	20.7%	20.7%
24	20.0%	20.0%

Now price an option, expiring in six months, to receive the two-year contract and deliver the one-year, both at \$80: Example 36.5's exchange option, with both legs on the same curve. Its value is Margrabe's formula with the variance of $\ln(F_2/F_1)$ accumulated over the option's life. In the two-factor model the long-term factor moves both contracts by the same amount and drops out of the ratio, so only the short-term factor contributes:

$$v^2 = \sigma_\chi^2 \left(e^{-\kappa T_1} - e^{-\kappa T_2}\right)^2 \frac{e^{2\kappa T_0} - 1}{2\kappa},$$

with $T_0 = 0.5$, $T_1 = 1$ and $T_2 = 2$. That gives $v = 5.92\%$ and, since the two forwards are equal, a value of $80\,[2N(v/2) - 1] = \$1.89$. The one-factor model has the same formula with σ and a in place of σ_χ and κ, and gives $v = 0.50\%$ and \$0.16. Both values are forward values, undiscounted.

The one-factor model misses because of how it had to be fitted. To carry a 20% volatility out to two years it must mean-revert slowly, so the two legs have almost the same volatility; being perfectly correlated as well, they move almost in step, and the spread barely moves. The two-factor model carries the long end with ξ and leaves χ free to revert fast, and it is the fast factor that moves one contract against the other. The two legs' correlation in the two-factor model is still 0.974, and the small gap from one carries all of the option's value.

And the misfit shows in the table. A desk that fitted it to front-month options instead would get the opposite error at the back. No single σ and a fit both ends of the curve, and that is the practical sign that a second factor is needed.

And add jumps, for power and increasingly for gas. Example 36.3 showed what omitting them costs.

Contracts on the forward curve are modeled on the curve itself. A model that gives each delivery period's forward its own volatility and its correlation with the others (Chapter 32's HJM approach, applied to commodity forwards) prices block forwards and the options on them consistently with the quoted curve. Spot models with spikes and forward-curve models are therefore complements rather than rivals: the first prices what settles on a day, the second what settles on a period, and Example 36.4 shows that a spot model calibrated to daily options leaves the value of a monthly contract open by a factor of three.

Commodity modeling is less standardized than rates or equities, because the physical constraints differ by commodity and a model that fits crude does not fit power. The practical discipline is to name the physical feature each model term is representing (this factor is inventory, this jump is a capacity constraint), because a commodity model with terms that correspond to

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nothing physical will be calibrated to fit and will fail out of sample.

36.9 BASIS: GRADE, LOCATION, AND TIMING

Section 3.3 defined basis risk generally. In commodities it is not a residual but the main event, because a hedger essentially never owns the contract grade at the contract location on the contract date.

Example 36.8 What the hedge leaves behind

A producer's realized price against a light sweet crude contract:

Contract (light sweet, Cushing, December)	\$78.00/bbl
Grade adjustment (heavier, higher sulfur)	−3.20
Location adjustment (wellhead to Cushing)	−2.40
Timing adjustment (November production)	−0.35
Realized price	**\$72.05/bbl**
Total basis	−\$5.95 **(7.6%)**

Call the futures price itself the **flat price**, to distinguish it from the basis. At a 32% volatility of the flat price and a 28% volatility *of the basis*:

Unhedged standard deviation of the realized price	\$24.96/bbl
Hedged standard deviation (basis only)	\$1.67/bbl
Risk reduction	**93.3%**

The hedge works: it removes 93% of the risk, which is why producers hedge.

But the residual is real and it is different in kind. Flat price is liquid, continuously quoted, and hedgeable; *basis is none of those*. Grade differentials move with refinery configuration and with regulation on sulfur content; location differentials move with pipeline capacity and can jump when a line is constrained; timing basis moves with local storage.

Three consequences follow.

Basis is hedgeable in some markets and not others. Where basis swaps trade (major US gas hubs, some crude grades), the residual can be reduced further. Where they do not, a producer's remaining exposure is to a variable with no market and no forecast.

Basis risk is worst exactly when flat price risk is worst. A pipeline constraint that sharply widens a location differential typically coincides with the demand event that moved the flat price. This is the wrong-way structure of Section 27.9 in a physical market, and it means the 93% figure overstates the protection in precisely the episodes the hedge was bought for.

And a hedger who reports the position as "hedged" has renamed the exposure rather than removed it: flat price is hedged and basis is not, and the two should be reported separately, for the reason Section 9.11 kept three valuation numbers apart.

36.10 STACK AND ROLL

The most instructive hedging failure in commodities is not a model error but a hedge that was economically correct and cash-flow wrong, and nothing in the pricing gives it away.

Example 36.9 A hedge that is right and still fails

A supplier sells ten years of fixed-price delivery, one million barrels a year. The far-dated futures are illiquid, so it hedges by **stacking** the whole ten-year exposure in the front month and rolling monthly.

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Total commitment	10,000,000 bbl
Stacked in the front month	10,000,000 bbl
Exposure arising in year one	1,000,000 bbl
Stack ratio	**10×**

Three problems follow, and each is separately sufficient.

One: the roll costs money, on the stacked notional. Section 5.11's roll yield applies ten times over:

Monthly contango	Annualized	Cost per year on a $780m stack
0.5%	6.2%	$48.1m
1.0%	12.7%	$98.9m
2.0%	26.8%	$209.2m

Two: the margin timing. A price fall makes the futures lose cash immediately while the offsetting gain on the supply contract accrues over a decade. The present value discounts each year's gain from mid-year at 5%, continuously compounded:

Price fall	Futures loss (cash, now)	PV of contract gain
−10%	$ 78.0m	$ 61.4m
−20%	$156.0m	$122.7m
−30%	$234.0m	$184.1m
−40%	$312.0m	$245.5m

A hedge that is economically right and cash-flow wrong can end a firm, which is Section 8.9's margin-as-liquidity-risk in its purest form and the mechanism behind more than one famous episode.

Three: the stack ratio is wrong even in principle. Two separate reasons apply, and they compound.

Discounting. At −10% the PV of the gain is $61.4m against a $78.0m loss, or **79%**. A one-for-one stack over-hedges in present-value terms by construction, because ten years of future gains are worth less than an immediate loss of the same nominal size.

The Samuelson maturity effect adds to it. Example 36.2 showed the front contract is roughly four times as volatile as the twelve-month at $a = 1.5$. Hedging a ten-year exposure with front-month futures therefore over-hedges the volatility as well.

The remedy is a maturity-weighted stack (fewer front contracts, weighted by e^{-aT} and by the discount factor), and the correct hedge ratio requires the mean-reversion parameter, which cannot be estimated reliably: Example 36.7 shows that no single a fits both ends of the curve, so the value obtained depends on which end it is fitted to. So the choice is between a hedge that is knowably wrong and one that is wrong by an unknown amount, and the first is preferable only if its error is reported.

The same mismatch returned in 2022, one generation after Metallgesellschaft (Section 3.10): European generators that had sold their output forward owed variation margin in cash as gas and power prices rose several-fold, while the offsetting gain on their plants would arrive only on delivery. Example 8.9 works the arithmetic and Section 40.12 tells the episode.

36.11 FINANCIAL PARTICIPATION AND THE CURVE

Commodity index investment (long-only, front-month, rolled mechanically) grew large enough to change the market it was invested in, by Section 22.8's mechanism. A published roll schedule creates a predictable flow: index funds sell the expiring contract and buy the next over a known window every month. A flow that is a published function of the calendar is a flow that can be anticipated, and the roll's cost to investors is the counterpart of someone else's return.

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And the aggregate position changed the curve's shape. Persistent long-only demand in the front months contributed to periods of contango in markets that had historically been backwardated, which raised the roll cost and, in turn, made the index products perform worse than the spot returns they were sold on.

The general lesson has now appeared four times in this book: portfolio insurance in Chapter 18, inverse volatility products in Section 22.8, snowball hedging in Chapter 35, and commodity index rolls here. A strategy whose trades are a published function of an observable is a strategy whose trades other people can price, and the resulting cost shows up as underperformance rather than as a loss anyone can point to.

36.12 FREIGHT, METALS, AND OTHER MARKETS

Three markets round out the picture, and each illustrates a point the chapter has already made.

Freight derivatives settle against published route indices: forward freight agreements on the cost of moving a cargo between two ports. Freight is the extreme case of non-storability: a voyage not made today cannot be banked, so the spot rate is set entirely by the balance of available tonnage and cargoes, and the market behaves more like power than like oil. Its economic role is to complete the locational spread of Section 36.7: an LNG cargo's destination option is only monetizable if the shipping leg can be priced, and freight is what prices it.

Base metals trade on a structure inherited from a physical warehousing system: delivery on individual days, the *prompt dates*, rather than in monthly contracts, and delivery against warrants, the title documents to metal in approved warehouses. The warehouse queue is a real economic variable: metal in a warehouse with a long load-out queue is not immediately available, so it supports the convenience yield of Section 36.1 even though the inventory statistics count it.

And a short-squeeze in a metal is the one place where the missing arbitrage of Section 36.1 becomes acute: a party short a physical contract cannot manufacture the metal, so the backwardation has no ceiling. The 2022 nickel episode is treated in Section 2.9 as a clearing and market-structure case; the price behavior that produced it is this chapter's asymmetry taken to its limit.

Agricultural markets add the feature that supply is annual and known only after harvest, so the curve breaks at the crop year. A calendar spread across a crop-year boundary is a different instrument from one within a year. It is the same warning as Section 36.2's, with a sharper discontinuity, because old-crop and new-crop are physically distinct goods.

The common thread is that in each market the financial contract is anchored to a physical constraint (a ship, a warehouse queue, a harvest), and the constraint, not the contract, sets the behavior the model has to reproduce.

36.13 EMISSIONS ALLOWANCES

A regulator caps emissions, issues allowances equal to the cap, and requires one surrendered per metric ton emitted; allowances trade, so *the price is the marginal cost of abatement*, the cost of the last ton of emissions cut to meet the cap. Three properties are all that matter here: they are **bankable**, so an allowance not surrendered this year can be kept for a later one; holding one is a registry entry; and borrowing one is hard, since the lending market is thin and holders with a compliance need will not lend.

So $u = 0$ and, absent doubt about future validity, $y = 0$, and (36.1) collapses to $F = Se^{r(T-t)}$—*the limiting case of a storable commodity*, in which the carry relation binds tighter than for anything physical in this chapter.

Where allowances trade. Allowances are created by governments, so the primary market is an auction run by the regulator or on its behalf, and the secondary market is mostly in futures

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rather than in the allowances themselves. A futures contract delivers allowances into the buyer's registry account at expiration, and most volume sits in each year's December contract, which lines up with the annual surrender deadline. The main schemes:

Scheme	Unit	Where it trades	Main users
EU Emissions Trading System (from 2005)	EU allowance (EUA)	Auctions on the European Energy Exchange (EEX); futures and options on ICE and EEX	Power generators; steel, cement, chemicals and refining; airlines on European routes; shipping from 2024
UK Emissions Trading Scheme (from 2021)	UK allowance (UKA)	Auctions and futures on ICE	Power generators, heavy industry, airlines on domestic routes
California and Québec cap-and-trade	California carbon allowance (CCA)	Joint quarterly auctions; futures on ICE and Nodal Exchange	Power generators and importers, large industry, suppliers of transport fuel and natural gas
Regional Greenhouse Gas Initiative (US Northeast)	RGGI allowance	Quarterly auctions; futures on ICE and Nodal Exchange	Fossil-fuel power generators only
China national emissions trading scheme (from 2021)	Chinese emission allowance (CEA)	Spot only, on the Shanghai Environment and Energy Exchange	Power generators; steel, cement and aluminum added from 2025

The EU market is the deepest, and the euro prices in this section's examples are of the order of an EU allowance. The last column lists the compliance entities, which hold allowances because they must surrender them. In every scheme they are joined by dealers who finance compliance purchases and funds that hold allowances as a view on the cap. From 2026 importers of steel, cement, aluminum, fertilizer and electricity into the EU must buy border-adjustment certificates priced off the allowance, a new class of buyer with a compliance horizon rather than a financial one. Section 36.18 returns to what each party wants.

Example 36.10 A storable asset with no storage cost

Spot €72.00, financing 3.5%. With $u = y = 0$, (36.1) gives

Years	Theoretical forward
0.5	€73.27
1	€74.56
2	€77.22
3	€79.97

The one-sided bound of Section 36.1 survives, and it points the same way. A curve *steeper* than carry is arbitraged (buy spot, bank, sell forward), and here the arbitrage is unusually clean, because banking is costless and unlimited, so the upper bound binds tightly. A curve flatter than carry cannot be arbitraged, since that requires borrowing an allowance.

The residual is therefore the whole content. Suppose the forward trades 3% below carry. Inverting (36.1) as in Example 36.1:

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Horizon	Implied y
0.5y	6.09%
1y	3.05%
2y	1.52%
3y	1.02%

A positive implied convenience yield here cannot be physical: there is no warehouse and no queue. It prices a probability that a banked allowance will not be usable as expected, and the $1/T$ decay in the table is the signature of a one-off event risk rather than a running yield: a constant y would not decay, while a single expected loss of value spread over a longer horizon does.

The same algebra that measured warehouse tightness in Example 36.1 is measuring something else entirely, and only the decay pattern distinguishes them.

Two things contaminate that residual before it can be used. Dealers finance compliance inventory (an installation buys allowances forward from a dealer holding the spot), so the curve prices a financing trade as much as an environmental scarcity and the residual is event risk *net of* a financing spread the market does not publish. And where banking stops at a phase boundary, the end of one of a scheme's multi-year trading periods, the allowance is a claim that expires, so the carry relation holds within a phase and not across one: Section 36.12's crop-year discontinuity with a legal calendar in place of a biological one, and a spread straddling the boundary is not a calendar spread.

Example 36.11 A convex supply curve meeting a vertical one

Abatement has a merit order, cheap measures first:

Cumulative abatement (Mt)	Marginal cost (€/t)
50	8
120	25
200	45
260	70
300	110
320	190

Supply is fixed, so a demand shift moves along this curve instead of being absorbed. Equal 20 Mt increases in required abatement:

Required abatement	Clearing price	Change
190 → 210 Mt	€42.50 → €49.17	+15.7%
250 → 270 Mt	€65.83 → €80.00	+21.5%
270 → 290 Mt	€80.00 → €100.00	+25.0%
290 → 310 Mt	€100.00 → €150.00	**+50.0%**

The same 20 Mt shift moves the price 15.7% in one place and 50.0% in another, with nothing changed but the position on the curve.

Two consequences follow for anyone carrying the risk. *Volatility rises as the cap tightens*, because the curve steepens, so a maturing allowance market becomes more volatile rather than less, which is the reverse of what happens as a financial market deepens. And a value at risk, the loss that should be exceeded only with a small stated probability, estimated on a historical window understates the risk systematically, because the elasticity that generated the history is not the elasticity now in force. The correct state variable is the position on the abatement curve, not the price level, and a risk model that conditions on the second is fitting the wrong thing.

Two adjacent instruments are routinely priced off allowances, and each fails for a reason

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this book has already named. A *voluntary credit* claims a reduction against an unobservable counterfactual, so credits are not fungible and a standardized futures contract on them prices the worst deliverable. That is Chapter 6's cheapest-to-deliver problem with integrity in place of duration, and it is why the market stays bilateral. A *renewable certificate* carries an obligation proportional to sales rather than a cap, so supply grows with generation and Example 36.11's vertical supply does not apply; and where the obligation can be discharged at a fixed **buy-out** price, that price is a hard cap and the certificate is a capped payoff by construction. An option on it inherits the cap, which is Section 34.12's capped cliquet in another market: past a threshold the vega changes sign with no change in the contract.

36.14 OPTIONS ON AN UNDERLYING WITH A BIMODAL DENSITY

Allowance options are quoted in Black's model like every other commodity option, and the quotes need reading differently. The reason is distributional rather than institutional, so the treatment here applies to any underlying whose uncertainty is dominated by a scheduled binary event (an election, a court ruling, a trial readout), and carbon is simply the case where the structure is permanent.

Example 36.12 What a bimodal density does to the smile

Model the one-year allowance price as a mixture: with probability $p = 0.35$ a tightening takes the forward to €110, otherwise it stays at €45, and within each regime the price is lognormal with 25% volatility. The mixture forward is

$$F = 0.35 \times 110 + 0.65 \times 45 = €67.75$$

and a call is priced by conditioning:

$$c(K) = e^{-rT}\Big[\, p\, c^{\mathrm{B}}(F_h, K, \sigma_r) + (1-p)\, c^{\mathrm{B}}(F_l, K, \sigma_r)\,\Big] \tag{36.5}$$

with c^{B} Black's undiscounted call, $F_h = 110$ and $F_l = 45$ the two regimes' forwards, $\sigma_r = 25\%$ the volatility within each regime, and $r = 3.5\%$, Example 36.10's financing rate. Inverting each price for a single Black volatility gives

Strike	Call	Implied vol	Delta error
€40	28.167	45.5%	−0.036
€50	21.934	51.1%	−0.069
€60	17.418	55.0%	−0.059
€67.75	14.535	**56.4%**	−0.026
€80	10.552	56.3%	+0.034
€90	7.769	54.7%	+0.067
€100	5.474	52.6%	+0.078
€120	2.407	48.4%	+0.060

Delta error is the mixture's true delta minus the Black delta computed at the fitted implied volatility.

Three results follow.

The implied volatility humps instead of smiling. It peaks at the money at 56.4% and falls to 45.5% and 48.4% in the wings—*the opposite shape to an equity index*. The density has two modes and a valley between them, so an at-the-money option straddles the valley and is expensive, while the wings have less mass than a lognormal of the same variance and are cheap. A desk that fits a standard skew parametrization to these quotes will fit the wrong sign and will extrapolate in the wrong direction.

The at-the-money quote understates the total uncertainty. The mixture's standard deviation is €36.32, which a lognormal matched on variance would report as **50.3%**—but that is not the

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at-the-money implied volatility, which is 56.4%, and neither number is the volatility of anything. Decomposing the variance: the regime uncertainty contributes 73% and the within-regime diffusion 27%, so nearly three-quarters of what a model calls volatility is not diffusion at all.

And the fitted volatility gives the wrong hedge. The delta error reaches −0.069 at €50 and **+0.078** at €100, eight delta points on a position that the trader believes is hedged. The sign flips across the strikes, so errors at different strikes offset only by accident: the book's net error depends on its strike mix, and it accumulates a residual that looks like gamma P&L and is actually model error. The remedy is to hedge with the mixture's delta and use the implied volatility only as a quoting convention.

The right diagnostic is the density, not the volatility. Section 19.2's Breeden–Litzenberger construction recovers Figure 36.2's left panel directly from the option prices and displays the bimodality that any single volatility number averages away.

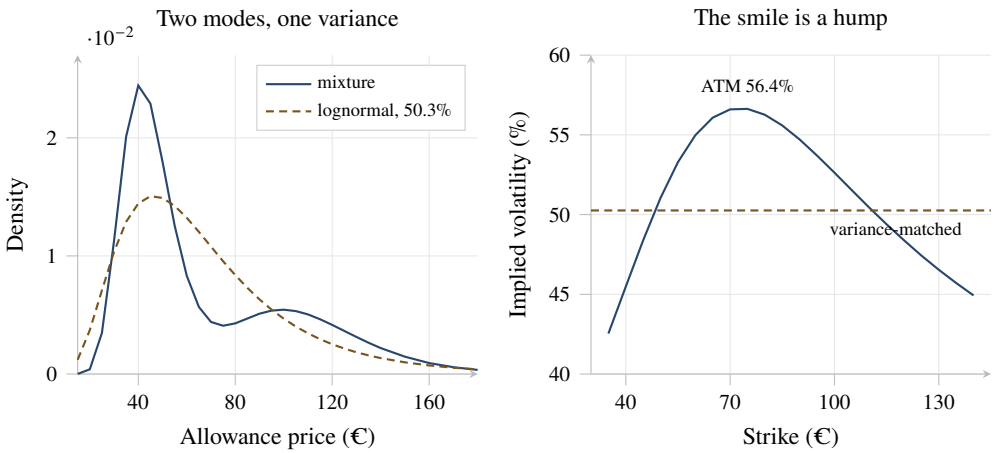


Figure 36.2: Example 36.12: mixture of lognormals, $p = 0.35$ on €110 against €45, 25% within each regime.

Figure 36.2 puts the two results side by side. The densities in the left panel have identical means and identical variances, and the lognormal's mode nearly coincides with the mixture's lower peak while standing only 61% as tall. From there it decays monotonically, *filling the valley the mixture leaves empty* (0.0095 against 0.0041 at €75) *and missing the upper mode entirely*. The right panel is what that misfit looks like through the only lens the market quotes in, and the hump is the diagnostic: an implied volatility surface that peaks at the money and falls in both wings is reporting a bimodal density, not a volatility.

36.15 WEATHER DERIVATIVES

Everything so far in this book has an underlying that can be held. *A weather index cannot be*, and the consequences for pricing are complete rather than partial.

A weather derivative settles against a measured index, most commonly **heating degree days** (HDD), HDD $= \max(65°\text{F} - T_{\text{avg}}, 0)$ summed over a period, with T_{avg} the day's average temperature, or its cooling counterpart, cooling degree days (CDD), which sum $\max(T_{\text{avg}} - 65°\text{F}, 0)$ instead. *The exposure it hedges is volume, not price*: a gas utility sells less gas in a warm winter, and no commodity hedge addresses that, because the utility's problem is that it has nothing to sell rather than that the price is wrong.

The index is a sum of many bounded daily quantities, so it is close to normal rather than lognormal, and the Bachelier form is the appropriate one. With index mean μ, standard

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deviation σ, strike K, tick value τ (the payment per index point) and $d = (\mu - K)/\sigma$,

$$P = \tau\big[\sigma\,\varphi(d) - (\mu - K)\,\Phi(-d)\big], \qquad C = \tau\big[\sigma\,\varphi(d) + (\mu - K)\,\Phi(d)\big] \tag{36.6}$$

with φ and Φ the standard normal density and distribution. There is no discounting of a forward and no volatility input in the usual sense: σ is the standard deviation of an index, estimated from history.

Example 36.13 Pricing without a hedge

November–March HDD at a reference station: mean 1200, standard deviation 140. A utility buys an HDD put struck at 1100 with a \$5,000 tick. Here $d = 100/140 = 0.7143$, and (36.6) gives

$P(\text{HDD} < 1100) = \Phi(-d)$	23.8%
$E[\text{shortfall} \mid \text{below}]$	82.2 HDD
Value of the put	**\$97,618**
Value of the matching call	\$597,618

Parity checks the pair: $C - P = \tau(\mu - K) = \$500{,}000$ exactly, which is the Bachelier analog of forward–strike parity and holds with no discounting because the index is not an asset.

That \$97,618 is an actuarial expectation, not a replication cost. No one can buy a degree day, so the writer cannot delta-hedge and the price is the expected payout plus a risk load set by capital and competition. Weather derivatives are priced like insurance and traded like derivatives, and every disagreement about value in this market is a disagreement about the distribution rather than about the model.

That makes the estimate of σ, and of μ, the entire pricing problem. Two conventions compete. *Burn analysis* takes the empirical distribution of the last 20 to 40 years of the index and averages the payoff; *index modeling* fits a daily temperature process and simulates. The first is transparent and assumes stationarity; the second imposes structure and can be wrong in ways burn analysis cannot. *And a warming trend makes stationarity false*, so detrending is a pricing input rather than a refinement: a 30-year window with a 1 HDD per year trend misstates μ by roughly 15 HDD. At this tick that is \$75,000 on a contract linear in the index, and it moves this put's value by about \$19,000, a fifth of its premium.

The geographic basis is the other half of the problem. The index is measured at one station; the utility's load is spread over a service area. Hedging load exposure L with the station index S, the minimum-variance ratio is $h^* = \rho\,\sigma_L/\sigma_S$ and the residual standard deviation is $\sigma_L\sqrt{1-\rho^2}$:

Station–load ρ	Variance reduction	Residual s.d.
0.95	90%	31%
0.90	81%	44%
0.85	**72%**	**53%**
0.70	49%	71%
0.50	25%	87%

At $\rho = 0.85$ the hedge removes 72% of the variance and leaves 53% of the standard deviation, and, unlike Section 36.9's commodity basis, *there is no instrument to hedge the remainder*, because a contract on the utility's own service area would have no counterparty willing to price it.

That asymmetry sets the market's structure. Weather risk is diversifiable across regions and perils but not hedgeable within one, so the natural writer is a party holding many uncorrelated exposures (a reinsurer or a diversified fund), and the market's participants look accordingly more like insurance than like a derivatives desk.

36.16 CATASTROPHE BONDS

The same economics can be funded. A catastrophe bond pays a coupon and returns principal unless a defined event occurs, in which case principal is written down and paid to the sponsor,

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the insurer or reinsurer that issued the bond to transfer the risk. Structurally it is Section 25.14's funded credit protection with a physical trigger, and Part V's machinery applies to the note's mechanics unchanged.

Pricing is an expected loss plus a load, and the load is most of it. The market quotes the ratio of spread to modeled expected loss as the **multiple**:

Expected loss	Spread	Multiple	Risk load
1.00%	4.50%	4.50×	3.50%
2.00%	6.50%	3.25×	4.50%
4.00%	10.00%	2.50×	6.00%
8.00%	16.00%	2.00×	8.00%

The multiple falls as the expected loss rises, which is the same pattern as credit: remote risks carry proportionally more compensation because the investor is being paid for the uncertainty in the loss estimate as well as for the loss. And the loss estimate comes from a vendor catastrophe model rather than from a market, so the multiple is partly a discount for model risk, which is why it widens after an event that the models did not anticipate, with no change in the physical hazard.

The trigger is the design decision and the trade-off is unavoidable. An **indemnity** trigger pays on the sponsor's actual losses, which is perfect for the sponsor and gives the investor exposure to the sponsor's underwriting quality. That is Section 26.9's incentive problem exactly. A **parametric** trigger pays on a measured physical quantity, wind speed or ground acceleration at a location, which removes the incentive problem and introduces basis risk of the kind Example 36.13 quantified. A trigger that eliminates basis risk transfers the moral hazard and one that eliminates the moral hazard transfers the basis risk, and no third option exists.

The diversification claim is unusually solid. Catastrophe risk is close to uncorrelated with financial markets because the mechanism is physical rather than statistical, which distinguishes it from almost every other diversification claim in this book, where the correlation is estimated from a sample and fails in the state where it is needed.

36.17 WHAT IS PRICED BY ARBITRAGE AND WHAT IS PRICED BY EXPECTATION

The instruments of the last four sections sort by how much of the book applies, and the sort matters because the formulas look identical in all three cases.

Allowance forwards: the machinery applies. Banking enforces the carry relation in one direction, and Example 36.10's residual is interpretable with the tools of Section 36.1.

Allowance options: the mechanics apply and the assumptions do not. (36.5) prices them correctly and Black's formula reports the answer in units of a volatility that is not one. The hedge ratio, not the price, is where the misreading costs money: Example 36.12's 0.078.

Weather and catastrophe: almost none of it applies. No replication, no arbitrage bound, no hedge for the writer. These are priced actuarially, and the derivatives contribution is contract design (the index, the trigger, the settlement) rather than the valuation.

The valuation machinery of Parts II and III derives a price from the cost of a hedge, and it returns a meaningful number only when the hedge exists. Where it does not, the same formula still returns a number, and that number is an expectation under an assumed distribution with nothing enforcing it. Both are legitimate quantities; confusing them is the error, and it is easier to make here than anywhere else in the book precisely because the arithmetic is unchanged.

One practical marker separates them. If the model's output changes when you change the hedging instrument, the price is a replication cost; if it changes only when you change the

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distribution, it is an expectation. A desk that cannot say which of the two it is quoting does not know what its risk report means.

36.18 WHO TRADES THESE MARKETS, AND WHY

Producers hedge to protect a capital-intensive investment made years before the revenue arrives. A producer's hedge is usually a term structure position, not a spot one, because the exposure is a production profile.

Consumers (airlines, utilities, food processors) hedge input costs, and their exposure is an average, which is why Section 33.8's Asian settlement is the norm.

Merchants and physical traders are the market's distinctive participant and have no analog in financial markets. They own the optionality of Section 36.7: storage, shipping, blending, and the right to deliver in one place rather than another. Their profit is the extrinsic value that the physical owner did not price, and it is earned by continuous re-optimization.

Utilities and generators are long the spreads of Section 36.6, whether or not they hedge them, and a generator that does not model its plant as a portfolio of options holds one anyway. In a market with an allowance price they are long the *clean* spread (power less fuel less carbon), so Section 36.6's exchange option becomes a three-asset spread requiring a correlation matrix rather than a single ρ, and the carbon leg is the one whose correlation is least stable.

Compliance buyers are the structural demand in allowances, and their hedging horizon is set by a surrender calendar rather than by a business plan, which makes their demand inelastic on known dates—an unusual and exploitable feature, and the reason well-designed schemes stagger the deadlines.

Utilities again, and reinsurers, are the two sides of weather. The absence of a natural opposite side within a region is why this market stayed small, and the point generalizes: a derivatives market needs not a risk but two parties with opposite exposures to it, which is what Section 33.10 established about inflation in a different setting.

Financial investors seek diversification and inflation sensitivity, and Section 36.11 describes what their participation did to the curve. In catastrophe risk their diversification claim is unusually well founded; in allowances their position is a directional bet on scarcity described as a commodity trade.

The recurring theme is that these markets are unusually close to something outside finance. The convenience yield is a fact about warehouses, the Samuelson maturity effect is a fact about how fast a shortage resolves, a negative power price is a fact about the cost of shutting down a turbine, and a carbon implied volatility is a fact about a cost curve. A model that loses sight of that will fit the data and misprice the constraint.

36.19 WHERE THIS CAME FROM

This subject is older than the rest of the book and grew from the study of storage rather than of options. Keynes (1930) argued that the futures price sits below the expected spot because hedgers pay speculators to bear risk, the normal backwardation of Section 5.13. Working (1949) and Brennan (1958) explained the shape of the curve instead by the cost and the benefit of holding the physical, which is where the convenience yield comes from.

The modern treatments make that yield stochastic, as Gibson and Schwartz (1990) did, and then ask how many factors a given commodity actually needs. *Schwartz (1997) is the standard comparison* and remains the natural starting point for choosing between one-, two- and three-factor specifications, because it compares them on the same data rather than advocating one.

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SUMMARY

One asymmetry generates the chapter. The cash-and-carry arbitrage runs one way (buy and store, never short and un-store), so contango is capped by storage economics and backwardation is not, and the cap disappears when storage fills. The convenience yield is the residual that makes the carry relation hold, and read off the curve it measures how scarce physical inventory is now and how long the scarcity is expected to last.

Storage also sets the volatility structure. Mean reversion makes near contracts more volatile than far ones, so options on different delivery months describe different markets and cannot be compared at one volatility. Electricity, which cannot be stored at all, is the extreme case: its forward is an expectation rather than an arbitrage price, its spot price spikes, and a lognormal matched on mean and variance misprices caps in both directions, overpricing those near the money and badly underpricing those far out. Which model a power contract needs depends on what it settles on. A daily cap depends only on each day's price; a contract on a monthly average depends on how spikes cluster in time, which no model of a single day can see.

Physical assets hide options. A power plant is a strip of exchange options whose value turns on the correlation between power and gas, and high correlation makes it cheap. Storage facilities, swing contracts and tolling agreements add flexibility, valued by dynamic programming because each choice uses up capacity that a later choice would have needed. Spreads between contract months need at least two factors, since a one-factor model fitted to a realistic curve moves the months in step and prices the spread close to zero. Hedging leaves a basis residual that is worst exactly when the hedge is needed, and a short-dated hedge of a long-dated commitment can be economically right and still fail for want of cash.

An emissions allowance is the limiting storable commodity. Banking is free, so the carry relation binds tightly, and whatever residual remains measures doubt about the scheme rather than physical scarcity. The price is set where a fixed supply meets a convex abatement curve, so volatility rises as the cap tightens. Policy events make the density of a future allowance price bimodal, so implied volatility humps instead of smiling, and the Black delta errs in a direction that flips across strikes.

At the far end the underlying cannot be held. Weather contracts and catastrophe bonds give the writer no hedge, so they are priced as an expectation plus a load for bearing the risk, and the buyer's residual exposure to its own location or portfolio cannot be removed. The distinction the chapter exists to make follows: if the output changes when you change the hedging instrument it is a replication cost, and if it changes only when you change the distribution it is an expectation.

FURTHER READING

Geman's *Commodities and Commodity Derivatives* is the standard reference, and the one that works through the physical constraints that generate the financial behavior. Clewlow and Strickland on energy derivatives is the practitioner's introduction to the modeling of Section 36.8 and is short. Schwartz and Smith's two-factor model is the paper behind that section's recommendation and is worth reading for the way it separates short-term and long-term uncertainty. On electricity specifically, Huisman and Mahieu's regime-switching model is the one behind (36.3), which is its one-day marginal; Cartea and Figueroa, and Geman and Roncoroni, add the mean reversion and jumps that Example 36.4 shows a monthly contract needs; and Weron's book surveys both families and the evidence that their fit varies from one market to the next. Eydeland and Wolyniec on energy and power risk management treats at length the spread and storage valuation of Sections 36.6 and 36.7. On the financialization question of Section 36.11, the empirical literature is contested and worth sampling from both sides; the disagreement is mostly about magnitude rather than direction. For Section 36.14, the mixture-of-lognormals treatment of event risk is developed in the literature on option prices

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around scheduled announcements. The carbon application is a special case of it, and reading the announcement papers first makes the hump in Figure 36.2 look inevitable rather than exotic. Weather derivatives have a small and stable literature of which the pricing material is mostly actuarial; the useful question, how to detrend, is treated mainly in the insurance journals rather than the finance ones. On catastrophe bonds the market's own offering documents set out the trigger trade-off of Section 36.16 and are worth reading in preference to summaries.

PRACTICE QUESTIONS

36.1 State the Samuelson relation and compute the twelve-month volatility at $\sigma_S = 50\%$ and $a = 1.0$. Using Example 36.2, state the one-month to twelve-month ratio at $a = 1.5$ and explain what a cross-maturity hedge ratio of one would cost.

36.2 Using Example 36.3, state the ratio of spike-model to lognormal cap values at \$200 and at \$400, and explain why calibrating the lognormal to at-the-money options does not close the gap.

36.3 Using Example 36.5, state the value of the option on one megawatt-hour at $\rho = 0.9$ and at $\rho = 0$, and explain why high power–gas correlation makes the spread option cheap rather than expensive.

36.4 Compute the 3-2-1 crack spread at crude \$80/bbl, gasoline \$2.50/gal, heating oil \$2.55/gal, and state what changes if the refinery's actual yield is 2-1-1.

36.5 Using Example 36.6, state the extrinsic share at 60% volatility and $\rho = 0.75$, and explain why the facility is usually leased by a trader rather than operated by the utility that built it.

36.6 Invert the carry relation to give the implied convenience yield, and compute it for $S = 80$, $F = 77$ at six months, $r = 4\%$, $u = 5\%$. State what a negative value would indicate and why it cannot be arbitraged away.

36.7 Using Example 36.8, state the risk reduction achieved by hedging the flat price, and explain why that figure overstates the protection in exactly the episodes the hedge was bought for.

36.8 Using Example 36.9, state the annual roll cost at 1% monthly contango on a \$780m stack, then state the cash loss and the present value of the offsetting gain at a 10% price fall and explain why the difference can end a firm.

36.9 Give the two independent reasons a one-for-one stack is the wrong size, state the remedy, and explain what the remedy requires that cannot be reliably estimated.

36.10 Compute the two-year allowance forward at a spot of €65 and a 4% rate, and state which side of that value is enforced by arbitrage.

36.11 Explain why a positive implied convenience yield in allowances cannot be physical, what its decay with maturity indicates, and how dealer financing of compliance inventory contaminates the reading.

36.12 Using Example 36.11, state the price change from two different 20 Mt shifts in required abatement, and explain what that implies for a value-at-risk estimated on a historical window.

36.13 Explain how a buy-out price caps a renewable certificate and what that does to the vega of an option written on it.

36.14 Using Example 36.12, state the at-the-money implied volatility and the two wing values, explain the shape, and state what a standard skew parametrization fitted to these quotes would get wrong.

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36.15 Explain why the variance-matched lognormal volatility of 50.3% differs from the 56.4% at-the-money implied volatility, and what each measures.

36.16 Using Example 36.12, state the largest delta error, explain why a book spread across strikes does not reliably net it out, and state what the residual will be mistaken for.

36.17 Explain why the Bachelier form rather than Black's is appropriate for a degree-day index, and use (36.6) to verify the put–call parity relation in Example 36.13.

36.18 Explain the difference between burn analysis and index modeling, state what each assumes, and quantify the effect of a 1 HDD-per-year trend on Example 36.13's contract.

36.19 Using Example 36.13, state the minimum-variance hedge ratio and the residual standard deviation at $\rho = 0.85$, and explain why the remainder cannot be hedged.

36.20 Explain why a catastrophe bond multiple falls as the expected loss rises, and why multiples widen after an unanticipated event with no change in the physical hazard.

FURTHER QUESTIONS

36.21 Example 36.2 gives the volatility term structure under mean reversion. Derive it from a mean-reverting spot process, then determine what the term structure looks like when storage is abundant, and use the result to explain why the observed term structure steepens when inventories fall.

36.22 Example 36.5 shows a 133% valuation error from ignoring correlation. Derive the spread option's sensitivity to ρ, determine where in the heat-rate spectrum the error is largest, and explain why a fleet of plants with different heat rates cannot be valued with a single correlation assumption.

36.23 Example 36.6 decomposes storage into intrinsic and extrinsic value. Formulate the storage operator's problem as a dynamic program with injection and withdrawal rate constraints, state how the constraints reduce the extrinsic value, and explain why a fast-cycling facility is worth more per unit of capacity than a slow one.

36.24 Example 36.8 finds a 93.3% risk reduction with a residual that is worst when the hedge is most needed. Model the basis as correlated with the flat price under stress, recompute the risk reduction conditional on a large flat-price move, and state how a producer should size the hedge given that conditional figure.

36.25 Example 36.10 reads the deviation from carry as a price of event risk and observes that it decays as $1/T$. Formalize this: model the allowance as paying off only if the scheme survives to T, derive the implied survival probability from the curve, state what else could produce the same discount, and propose a test on the term structure that would distinguish the explanations.

36.26 Example 36.12 produces a delta error that changes sign across strikes. Derive the mixture's delta in closed form from (36.5), determine the strike at which the error is largest as a function of p, and construct the portfolio of vanilla options that would hedge the regime exposure directly rather than through delta.

36.27 Example 36.13 finds 53% residual standard deviation at a 0.85 correlation. Determine how many independent station contracts would be needed to bring the residual below 20%, state the correlation structure your answer assumes, and explain why the market has not developed such multi-station products.

36.28 Example 36.4 holds each day's price distribution fixed and varies only how spikes cluster. Prove that the strip of daily caps cannot depend on the clustering, derive the variance of

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the monthly average as a function of the episode length, and explain why options on block futures trade at implied volatilities far below those of daily options in the same market.

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